Suggested Answers to Problem Set 2

1. (a) The AR polynomial is $\phi(z) = 1 - z$, which has root $z = 1$. The MA polynomial is $\theta(z) = 1 - \frac{1}{2}z - \frac{1}{2}z^2 = (1 - z)\left(1 + \frac{1}{2}z\right)$, which has roots $z_1 = 1$ and $z_2 = -2$. After removing the common root 1, the model reduces to $y_t = \varepsilon_t + \frac{1}{2}\varepsilon_{t-1}$. Therefore, this is an $MA(1)$ model, or $ARMA(0, 1)$ model with $p = 0$ and $q = 1$. It is stationary since all finite-order MA models are stationary. It is invertible since $z_2 = -2$ is outside the unit circle.

   (b) The AR polynomial is $\phi(z) = 1 - 2z + 2z^2$, which has roots $\frac{1 \pm i}{2}$. The MA polynomial is $\theta(z) = 1 - \frac{3}{4}z$, which has a root $\frac{4}{3}$. There are no common roots. The model is an $ARMA(2, 1)$ with $p = 2$ and $q = 1$. It is non-stationary since $\frac{1 \pm i}{2}$ lie inside the unit circle, but it is invertible since $\frac{4}{3}$ is outside the unit circle.

   (c) The AR polynomial is $\phi(z) = \left(1 + \frac{3}{4}z\right)(1 - 2z) = 0$. The two roots are $z_1 = -\frac{4}{3}$, $z_2 = \frac{1}{2}$. The MA polynomial is $\theta(z) = 1 - 2z$ with root $\frac{1}{2}$. There is a common root $\frac{1}{2}$, so the model is in fact an $AR(1)$ model, or $ARMA(1, 0)$ with $p = 1$ and $q = 0$, given by $y_t = -\frac{3}{4}y_{t-1} + \varepsilon_t$. It is clear that the model is both stationary and invertible.

2. First, multiplying both sides of the equation by y_t and taking expectations yield

$$\gamma_0 = \phi_1\gamma_1 + \phi_2\gamma_2 + E(\varepsilon_t y_t) + \theta_1 E(\varepsilon_{t-1} y_t). \tag{1}$$

   Note that

$$\begin{aligned} E(\varepsilon_{t-1} y_t) &= E\left[\varepsilon_{t-1}\left(\phi_1 y_{t-1} + \phi_2 y_{t-2} + \varepsilon_t + \theta_1\varepsilon_{t-1}\right)\right] \\ &= \phi_1 E(\varepsilon_{t-1} y_{t-1}) + \theta_1 E\left(\varepsilon_{t-1}^2\right) \\ &= (\phi_1 + \theta_1)\sigma^2. \end{aligned}$$

   Using the above and $E(\varepsilon_t y_t) = \sigma^2$ in (1) gives

$$\begin{aligned} \gamma_0 &= \phi_1\gamma_1 + \phi_2\gamma_2 + \sigma^2 + \theta_1(\phi_1 + \theta_1)\sigma^2 \\ &= \phi_1\gamma_1 + \phi_2\gamma_2 + \left(\theta_1\phi_1 + \theta_1^2 + 1\right)\sigma^2, \end{aligned}$$

   as desired. Next, multiplying both sides of the model by y_{t-1} and taking expectations yield

$$\begin{aligned} \gamma_1 &= \phi_1\gamma_0 + \phi_2\gamma_1 + E(\varepsilon_t y_{t-1}) + \theta_1 E(\varepsilon_{t-1} y_{t-1}) \\ &= \phi_1\gamma_0 + \phi_2\gamma_1 + \theta_1\sigma^2. \end{aligned}$$

   The third equation can be readily shown by multiplying both sides of the model by y_{t-h}, for $h \geq 2$, and taking expectations.

3. (a) The 1-step-ahead forecast is

$$\hat{y}_{101|100} = 3 + 0.3y_{100} + 0.04y_{99} = 4.21,$$

with forecast error

$$\begin{aligned} e_{t+1|t} &= (3 + 0.3y_t + 0.04y_{t-1} + \varepsilon_{t+1}) - (3 + 0.3y_t + 0.04y_{t-1}) \\ &= \varepsilon_{t+1}. \end{aligned}$$

$Var\left(e_{101|100}\right) = Var\left(\varepsilon_{101}\right) = 0.25$, and it follows that $s.e.\left(e_{101|100}\right) = 0.5$. The 95% prediction interval is $4.21 \pm 1.96 \times 0.5 = [3.23, 5.19]$. The 2-step-ahead forecast is

$$\hat{y}_{102|100} = 3 + 0.3\hat{y}_{101} + 0.04y_{100} = 3 + 0.3 \times 4.21 + 0.04 \times 3.5 = 4.403,$$

with forecast error

$$\begin{aligned} e_{t+2|t} &= (3 + 0.3y_{t+1} + 0.04y_t + \varepsilon_{t+2}) - (3 + 0.3\hat{y}_{t+1} + 0.04y_t) \\ &= \varepsilon_{t+2} + 0.3\varepsilon_{t+1}. \end{aligned}$$

The variance of the 2-step-ahead forecast error is

$$Var\left(e_{102|100}\right) = (1 + 0.09) \times 0.25 = 0.2725$$

and hence $s.e.\left(e_{102|100}\right) = 0.522$. The 95% prediction interval is $4.403 \pm 1.96 \times 0.522 = [3.380, 5.426]$.

(b) This is a stationary process. The long-run forecast converges to the unconditional mean of y_t,

$$E\left(y_t\right) = \frac{3}{1 - 0.3 - 0.04} = 4.55,$$

with error variance equaling the unconditional variance of y_t,

$$Var\left(y_t\right) = \frac{0.25}{1 - 0.3^2 - 0.04^2} = 0.275,$$

and hence $s.e.\left(y_t\right) = 0.524$.

4. (a) The 1-step-ahead forecast is

$$\hat{y}_{T+1|T} = \phi_0 + \phi_1 y_T + \theta_1 \varepsilon_T.$$

(b) The 2-step-ahead forecast is

$$\begin{aligned}\hat{y}_{T+2|T} &= \phi_0 + \phi_1 \hat{y}_{T+1|T} \\ &= \phi_0 + \phi_1 \left(\phi_0 + \phi_1 y_T + \theta_1 \varepsilon_T\right) \\ &= \phi_0 \left(1 + \phi_1\right) + \phi_1^2 y_T + \phi_1 \theta_1 \varepsilon_T.\end{aligned}$$

It can be seen that the h-step-ahead forecast is (for $h \geq 2$)

$$\begin{aligned}\hat{y}_{T+h|T} &= \phi_0 + \phi_1 \hat{y}_{T+h-1|T} \\ &= \phi_0 \sum_{i=0}^{h-1} \phi_1^i + \phi_1^h y_T + \phi_1^{h-1} \theta_1 \varepsilon_T,\end{aligned}$$

which decays at the rate of ϕ_1 toward $E\left(y_t\right) = \phi_0 / \left(1 - \phi_1\right)$, as h tends to infinity, since $|\phi_1| < 1$ under stationarity.

(c) The 1-step-ahead forecast error is

$$\begin{aligned}e_{T+1|T} &= y_{T+1} - \hat{y}_{T+1|T} \\ &= \left(\phi_0 + \phi_1 y_T + \varepsilon_{T+1} + \theta_1 \varepsilon_T\right) - \left(\phi_0 + \phi_1 y_T + \theta_1 \varepsilon_T\right) \\ &= \varepsilon_{T+1},\end{aligned}$$

and its variance is $Var\left(e_{T+1|T}\right) = \sigma^2$. The 2-step-ahead forecast error is

$$\begin{aligned}e_{T+2|T} &= y_{T+2} - \hat{y}_{T+2|T} \\ &= \left(\phi_0 + \phi_1 y_{T+1} + \varepsilon_{T+2} + \theta_1 \varepsilon_{T+1}\right) - \left(\phi_0 + \phi_1 \hat{y}_{T+1|T}\right) \\ &= \phi_1 \hat{e}_{T+1|T} + \varepsilon_{T+2} + \theta_1 \varepsilon_{T+1} \\ &= \varepsilon_{T+2} + \left(\phi_1 + \theta_1\right) \varepsilon_{T+1},\end{aligned}$$

and its variance is

$$Var\left(e_{T+1|T}\right) = \left[1 + \left(\phi_1 + \theta_1\right)^2\right] \sigma^2.$$

The 3-step-ahead forecast error is

$$\begin{aligned}e_{T+3|T} &= y_{T+3} - \hat{y}_{T+3|T} \\ &= \left(\phi_0 + \phi_1 y_{T+2} + \varepsilon_{T+3} + \theta_1 \varepsilon_{T+2}\right) - \left(\phi_0 + \phi_1 \hat{y}_{T+2|T}\right) \\ &= \phi_1 \hat{e}_{T+2|T} + \varepsilon_{T+3} + \theta_1 \varepsilon_{T+2} \\ &= \varepsilon_{T+3} + \left(\phi_1 + \theta_1\right) \varepsilon_{T+2} + \phi_1 \left(\phi_1 + \theta_1\right) \varepsilon_{T+1},\end{aligned}$$

and its variance is

$$Var\left(e_{T+3|T}\right) = \left[1 + \left(1 + \phi_1^2\right) \left(\phi_1 + \theta_1\right)^2\right] \sigma^2.$$

It can be seen that for $h \geq 2$,

$$Var\left(e_{T+h|T}\right) = \left[1 + \left(\sum_{i=0}^{h-2} \phi_1^{2i}\right)(\phi_1 + \theta_1)^2\right]\sigma^2,$$

which grows with h and converges to

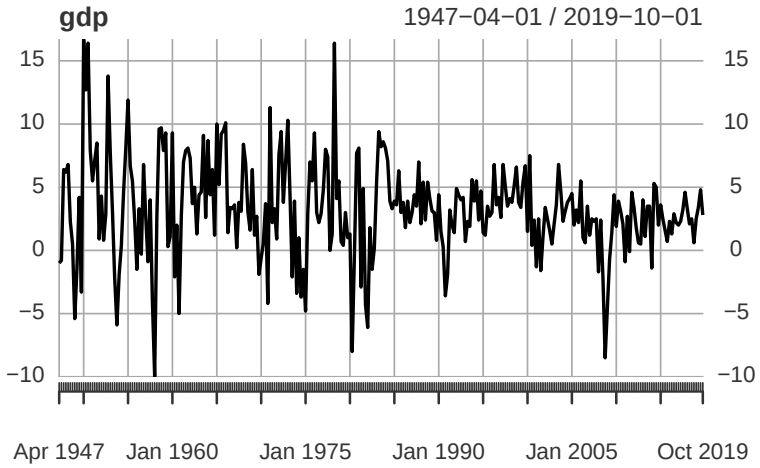
$$\left[1 + \frac{1}{1-\phi_1^2}(\phi_1 + \theta_1)^2\right]\sigma^2 = \frac{\left(1 + 2\phi_1\theta_1 + \theta_1^2\right)\sigma^2}{1-\phi_1^2}, \tag{2}$$

as $h \to \infty$. We have shown in class that (2) is the unconditional variance of the $ARMA\,(1,1)$ model.

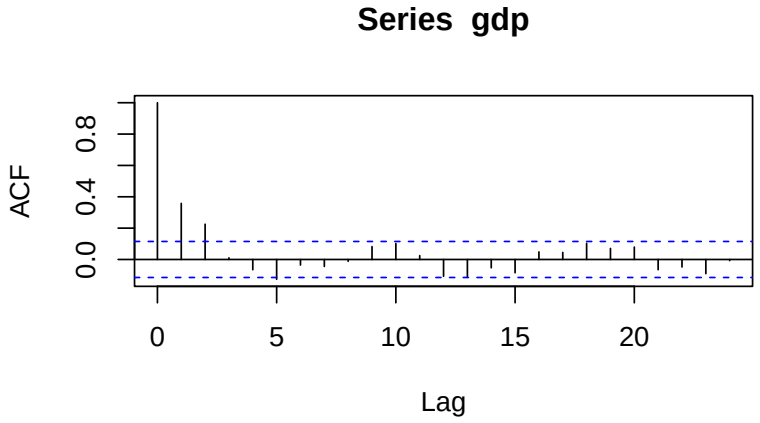
Question 5

(a) The plots show that the series seems stationary.

```
library(quantmod)
library(forecast)
gdp.full <- getSymbols("A191RL1Q225SBEA", src="FRED", auto.assign=FALSE)
gdp <- gdp.full["/2019"]
colnames(gdp) <- "gdp"
plot(gdp)
```




```
acf(gdp)
```



(b) I will use the default estimation method of `arima()` as an illustration. You are encouraged to experiment with different methods. First, you are asked to write down the estimated (or fitted) equation. This is to examine if you know how to read the outputs. As I explained in class, the estimated mean is called "intercept" in the `arima()` output, so you will need to calculate the intercept term for AR and ARMA models (the intercept equals mean for MA models). Let y_t denote the *gdp* series. According to the following outputs, we have

The fitted AR(1) model is $y_t = 2.0674 + 0.3584 y_{t-1} + \epsilon_t$, which can be written as $y_t - 3.2217 = 0.3583(y_{t-1} - 3.2217) + \epsilon_t$, where 3.2217 is the estimated mean.

The fitted MA(1) model is $y_t = 3.2268 + \epsilon_t + 0.2670\epsilon_{t-1}$.

The fitted ARMA(1,1) model is $y_t = 1.5801 + 0.5091 y_{t-1} + \epsilon_t - 0.1697 \epsilon_{t-1}$.

Diagnostic checks suggest that none of the three candidate models is adequate at the 5% level. The residuals of the three models do not behave like white noise as shown by the time plots, ACF plots and the Ljung-Box test results.

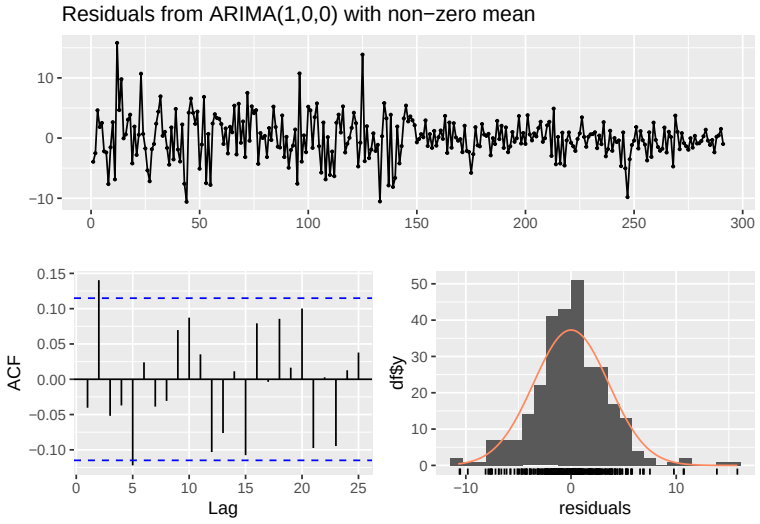
```
# AR(1)
fit.ar1 <- arima(gdp, order=c(1,0,0))
fit.ar1
```

```
##
## Call:
## arima(x = gdp, order = c(1, 0, 0))
##
## Coefficients:
##          ar1  intercept
##       0.3583     3.2217
## s.e.  0.0547     0.3255
##
## sigma^2 estimated as 12.75:  log likelihood = -783.31,  aic = 1572.61
```

```
fit.ar1$coef[2]*(1-fit.ar1$coef[1]) # calculate the intercept
```

```
## intercept
##  2.067418
```

```
checkresiduals(fit.ar1)
```

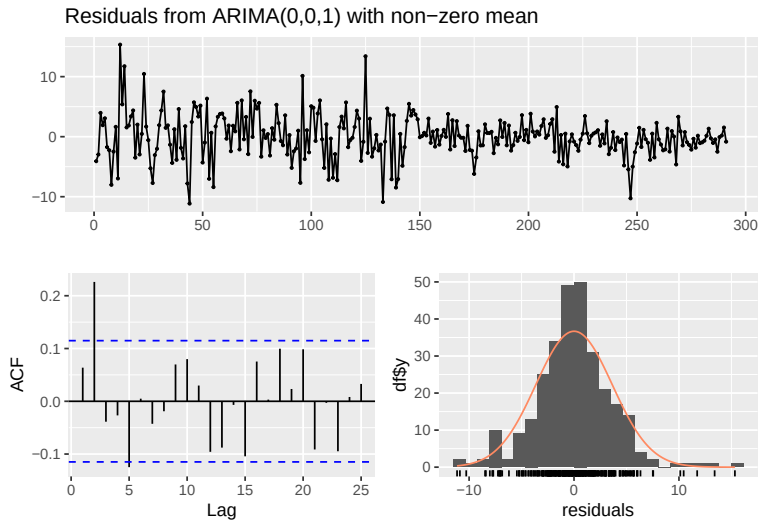



```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(1,0,0) with non-zero mean
## Q* = 16.625, df = 9, p-value = 0.05493
##
## Model df: 1.   Total lags used: 10
```

```r
# MA(1)
fit.ma1 <- arima(gdp, order=c(0,0,1))
fit.ma1  # intercept equals mean for MA models
```

```
##
## Call:
## arima(x = gdp, order = c(0, 0, 1))
##
## Coefficients:
##          ma1  intercept
##       0.2670     3.2268
## s.e.  0.0466     0.2700
##
## sigma^2 estimated as 13.23:  log likelihood = -788.72,  aic = 1583.44
```

```r
checkresiduals(fit.ma1)
```




```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(0,0,1) with non-zero mean
## Q* = 25.657, df = 9, p-value = 0.002324
##
## Model df: 1.   Total lags used: 10
```

```r
# ARMA(1,1)
fit.arma11 <- arima(gdp, order=c(1,0,1))
fit.arma11
```

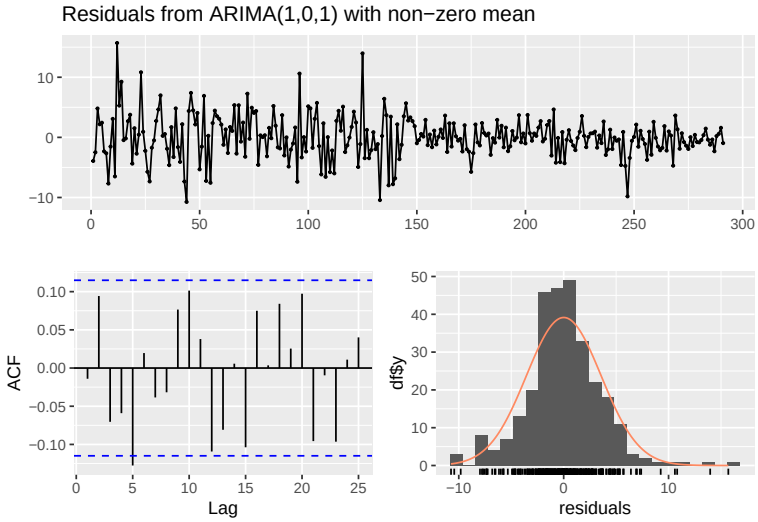
```
##
## Call:
## arima(x = gdp, order = c(1, 0, 1))
##
## Coefficients:
```

```
##          ar1      ma1  intercept
##       0.5091  -0.1697     3.2188
## s.e.  0.1044   0.1129     0.3517
##
## sigma^2 estimated as 12.65:  log likelihood = -782.27,  aic = 1572.53
```

```
fit.arma11$coef[3]*(1-fit.arma11$coef[1]) # calculate the intercept
```

```
## intercept
##  1.580067
```

```
checkresiduals(fit.arma11)
```




```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(1,0,1) with non-zero mean
## Q* = 15.791, df = 8, p-value = 0.04547
##
## Model df: 2.   Total lags used: 10
```

(c)

The PACF plot shows that the second and third lags are marginally significant. We may want to choose 3 to be the maximum lag order, or 4 to play it safe since we have quarterly data. Using the `auto.arima()` function, AR(3) is selected to be the best model based on both aicc and aic, whereas AR(1) is selected based on bic.

The residual diagnostics clearly show that AR(1) is not adequate and AR(3) seems to be a good fit.

You can also fit AR(1)-AR(4) models in turn and compare AIC/BIC. You will reach the same conclusion as using `auto.arima()`.

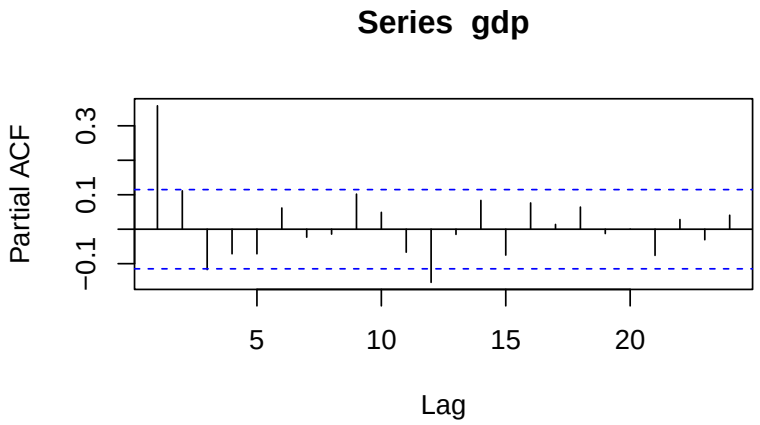
In conclusion, we have identified AR(3) to be the best model for GDP growth rates. Let $y_t = gdp_t - 3.2189$. The fitted model is

$$\hat{y}_t = \underset{(0.0581)}{0.3322} y_{t-1} + \underset{(0.0607)}{0.1506} y_{t-1} - \underset{(0.0583)}{0.1198} y_{t-2},$$

where standard errors are in parentheses.

Remark: We had not covered the unit root tests when you worked on this question, so the answer sets `d=0` in `auto.arima()` to restrict attention to stationary models. You are encouraged to experiment with unit root tests and seasonal ARIMA models, and compare your finding with the model using `auto.arima()` without restrictions.

```
pacf(gdp)
```




```
# To fit a pure AR model we need to change a few default arguments of auto.arima
auto.arima(gdp, max.p=4, max.q=0, d=0, seasonal=FALSE, ic="aicc")
```

```
## Series: gdp
## ARIMA(3,0,0) with non-zero mean
##
## Coefficients:
##          ar1     ar2      ar3    mean
##       0.3322  0.1506  -0.1198  3.2189
## s.e.  0.0581  0.0607   0.0583  0.3237
##
## sigma^2 = 12.57:  log likelihood = -779.36
## AIC=1568.73   AICc=1568.94   BIC=1587.09
```

```
auto.arima(gdp, max.p=4, max.q=0, d=0, seasonal=FALSE, ic="aic")
```

```
## Series: gdp
## ARIMA(3,0,0) with non-zero mean
##
## Coefficients:
##          ar1     ar2      ar3    mean
##       0.3322  0.1506  -0.1198  3.2189
## s.e.  0.0581  0.0607   0.0583  0.3237
##
## sigma^2 = 12.57:  log likelihood = -779.36
## AIC=1568.73   AICc=1568.94   BIC=1587.09
```

```
auto.arima(gdp, max.p=4, max.q=0, d=0, seasonal=FALSE, ic="bic")
```

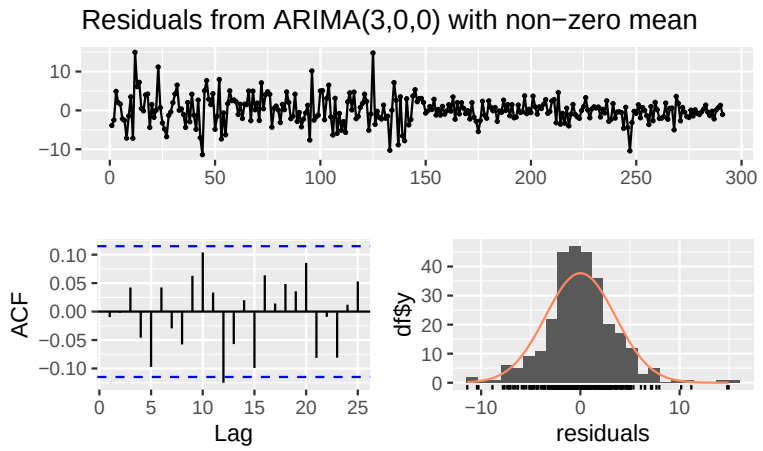
```
## Series: gdp
```

```
## ARIMA(1,0,0) with non-zero mean
##
## Coefficients:
##          ar1    mean
##       0.3583  3.2217
## s.e.  0.0547  0.3255
##
## sigma^2 = 12.83:  log likelihood = -783.31
## AIC=1572.61   AICc=1572.7   BIC=1583.63
```

```
fit.ar3 <- arima(gdp, order=c(3,0,0))
fit.ar3
```

```
##
## Call:
## arima(x = gdp, order = c(3, 0, 0))
##
## Coefficients:
##          ar1     ar2      ar3  intercept
##       0.3322  0.1506  -0.1198     3.2189
## s.e.  0.0581  0.0607   0.0583     0.3237
##
## sigma^2 estimated as 12.4:  log likelihood = -779.36,  aic = 1568.73
```

```
checkresiduals(fit.ar3)
```



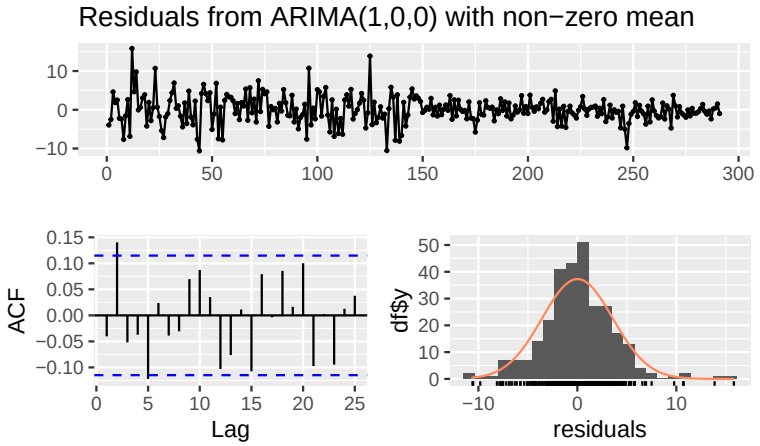

```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(3,0,0) with non-zero mean
## Q* = 10.285, df = 7, p-value = 0.173
##
## Model df: 3.   Total lags used: 10
```

```
fit.ar1 <- arima(gdp, order=c(1,0,0))
fit.ar1
```

```
##
```

```
## Call:
## arima(x = gdp, order = c(1, 0, 0))
##
## Coefficients:
##          ar1  intercept
##       0.3583     3.2217
## s.e.  0.0547     0.3255
##
## sigma^2 estimated as 12.75:  log likelihood = -783.31,  aic = 1572.61
```

```
checkresiduals(fit.ar1)
```




```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(1,0,0) with non-zero mean
## Q* = 16.625, df = 9, p-value = 0.05493
##
## Model df: 1.   Total lags used: 10
```

```
fit.ar4 <- arima(gdp, order=c(4,0,0))
fit.ar2 <- arima(gdp, order=c(2,0,0))
AIC(fit.ar1, fit.ar2, fit.ar3, fit.ar4)
```

```
##         df      AIC
## fit.ar1  3 1572.615
## fit.ar2  4 1570.918
## fit.ar3  5 1568.727
## fit.ar4  6 1569.154
```

```
BIC(fit.ar1, fit.ar2, fit.ar3, fit.ar4)
```

```
##         df      BIC
## fit.ar1  3 1583.635
## fit.ar2  4 1585.612
## fit.ar3  5 1587.093
## fit.ar4  6 1591.194
```

(d) Yes, there exists business cycles since the characteristic equation has complex roots, which are $0.409 \pm 0.282i$. The average length is about 10.4 quarters.

```
fit.ar3$coef
```

```
##        ar1        ar2        ar3  intercept
##  0.3321669  0.1506493 -0.1197688  3.2189141
```

```
(roots <- 1/polyroot(c(1, -fit.ar3$coef[1:3]))) # inverse AR roots (lambda)
```

```
## [1]  0.4089339-2.817139e-01i -0.4857008+6.620404e-19i  0.4089339+2.817139e-01i
```

```
2*pi/acos( Re(roots[1])/Mod(roots[1]) )
```

```
## [1] 10.41579
```

```
2*pi/Arg(roots[3]) # alternative approach
```

```
## [1] 10.41579
```

(e)

```
(yhat <- forecast(fit.ar3, h=4, level=95))
```

```
##     Point Forecast     Lo 95    Hi 95
## 292       3.296266 -3.606048 10.19858
## 293       2.992134 -4.281002 10.26527
## 294       3.205411 -4.287488 10.69831
## 295       3.171000 -4.322813 10.66481
```

(f) Note that you have to re-estimate the model using the subsample. You can verify that in this case AR(3) is still selected as the best model. While the 1-step-ahead forecast is quite close to the actual value, the other forecasts are a bit off. The 95% prediction intervals cover the actual value, but they are quite wide.

```
gdp_subsample <- gdp["/2018"]
fit_subsample <- arima(gdp_subsample, order=c(3,0,0))
yhat_last4 <- forecast(fit_subsample, h=4, level=95)
data.frame(tail(gdp,4), yhat_last4)
```

```
##            gdp Point.Forecast     Lo.95     Hi.95
## 2019-01-01 2.5       2.369185 -4.577781  9.316151
## 2019-04-01 3.4       2.622529 -4.698177  9.943236
## 2019-07-01 4.8       3.200479 -4.342396 10.743354
## 2019-10-01 2.8       3.220028 -4.323884 10.763939
```